Robot Safety & Trust

(Cornell University, 2022)

Group B: Human-Robot Interaction & Collaboration

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Intro to HRC & Importance of Trust

Gaurav

(Blue Frog Robotics, 2023)

Human-Robot Collaboration

It is everywhere - Blending high-level human intelligence with low level robot precision.

Elderly Healthcare
(Founderspace.com, 2025)

Medical Surgeries
(da Vinci Surgical System, 2000)

Advanced Manufacturing
(Boston Dynamic, 2025)

The vital ingredient is trust. The right level lets people skip needless checks, interruptions and corrections..

What Do We Mean by “Trust”?

> The human’s expectation that the robot will carry out tasks with the desired performance.

It captures a willingness to depend on the robot, a shared sense of its intent, and a positive expectation of the outcome.

With a high level of trust, people:

- Assign tasks freely
- Share plans openly
- Interrupt far less
- Collaborate with confidence

The Trust Calibration Problem

Both too much and too little trust damage collaboration.

Under-Trust

Human does it by hand, ignoring the robot.

(Reddit, 2026)

The Goal

Appropriate Trust

Confident, low-interruption teamwork — this is what we aim for.

Over-Trust

Blindly trusts the robot — it makes a mess.

(Imgur, 2021)

Mis-calibrated trust breaks collaboration in both directions.

Why Do Robot Make Mistake?

Real-world conditions cause errors that no robot can fully avoid.

Environmental Disturbance
(Lindleys Autocentres, 2019)

Sensor Uncertainty
(Adam Smith, 2017)

Planning Degradation
(Reddit, 2025)

In the paper, these surface as four mistake types:

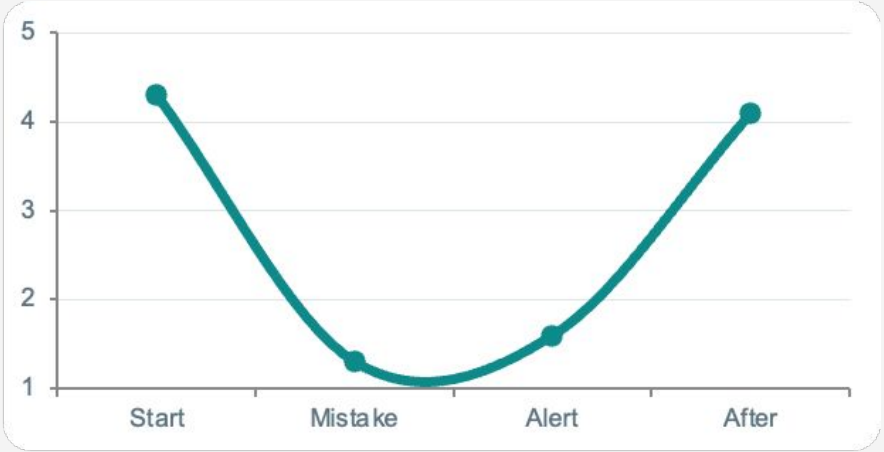
- Wrong action
- Wrong region
- Wrong pose
- Wrong spatial relation

The Fix: Attention Transfer

Core idea: promptly correcting a mistake restores human trust.

- Human spots the abnormal behavior
- Gives a verbal alert — “You’re releasing the cup!”
- H2R-AT fuses the words (LSTM) with what the robot sees (VGG)
- Robot locates the error and corrects it
- **Trust is repaired**

Trust across the interaction



Trust falls when the mistake appears, then climbs back once the robot fixes itself.

Overview

01

Vision For Incident Prevention

Panuwat

02

Verbal Communication For Correction

Alwin

03

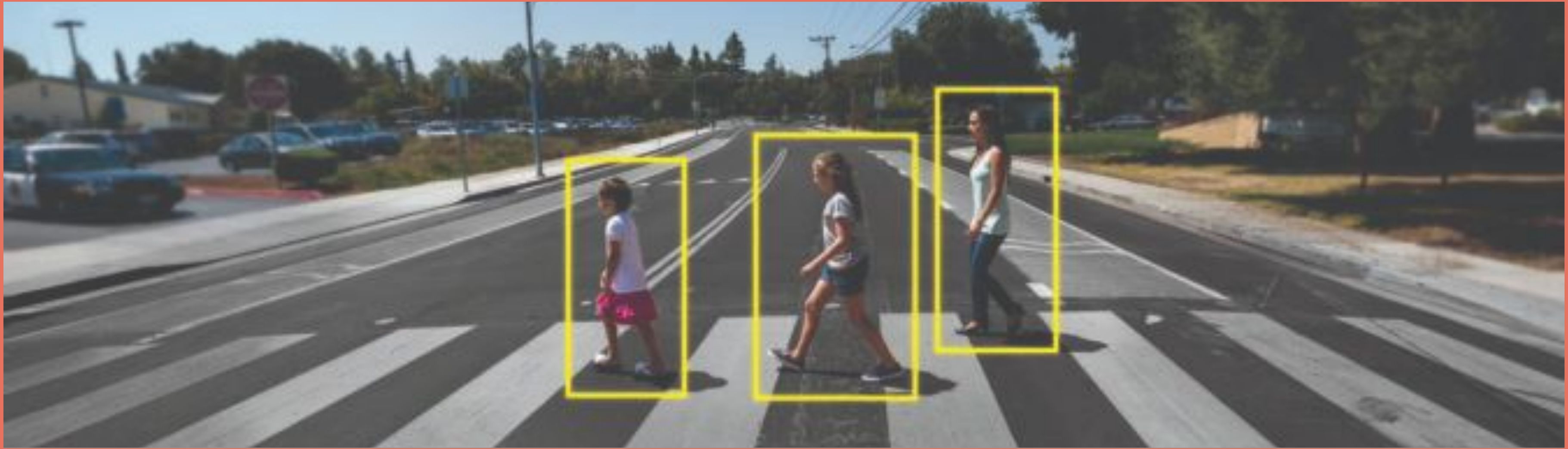
Haptic To Reduce Harm

Hunter

04

Combined Techniques To Increase **Trust**

張泳德



How **vision** is used for accidents prevention?

Panuwat

(Nvidia, 2019)

The Evolving Challenge of Safety

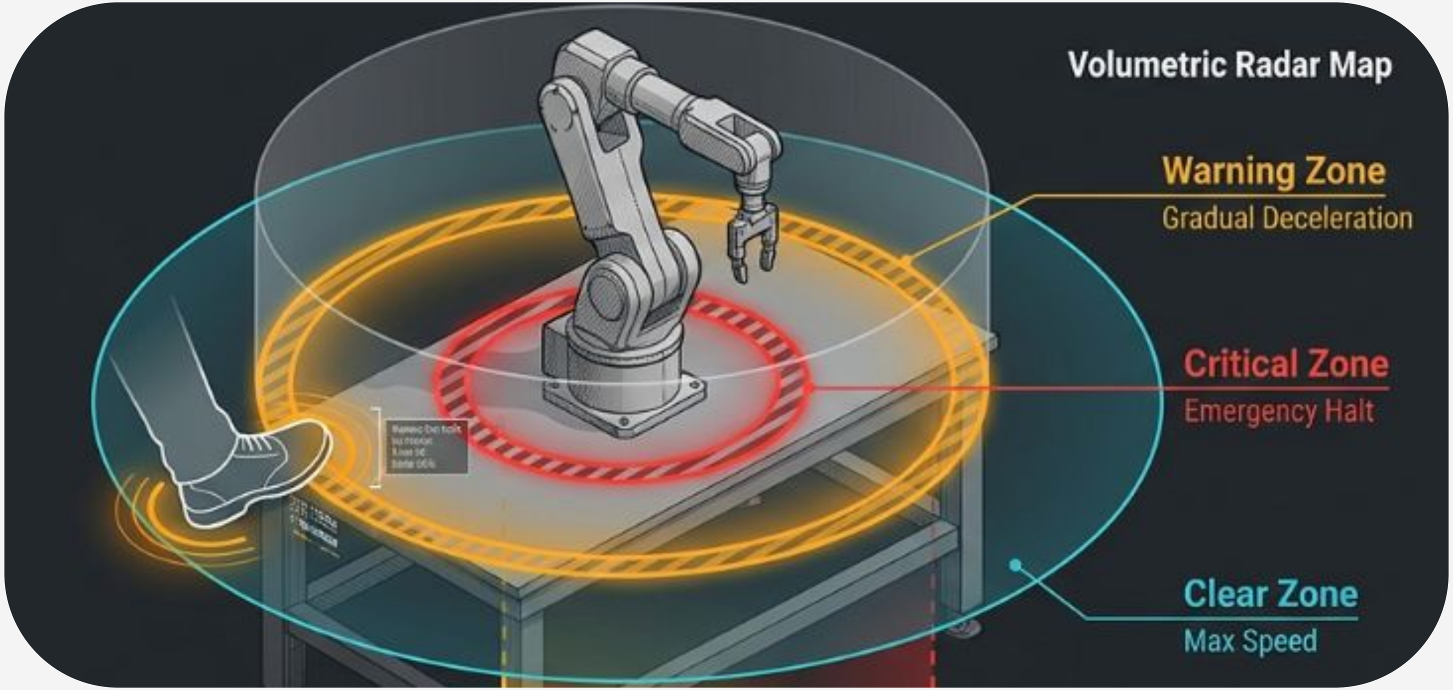
The Steel Cage (Past)

(Padget Technologies, 2020)

The Algorithm Cage (Present / Future)

(Universal Robots, 2020)

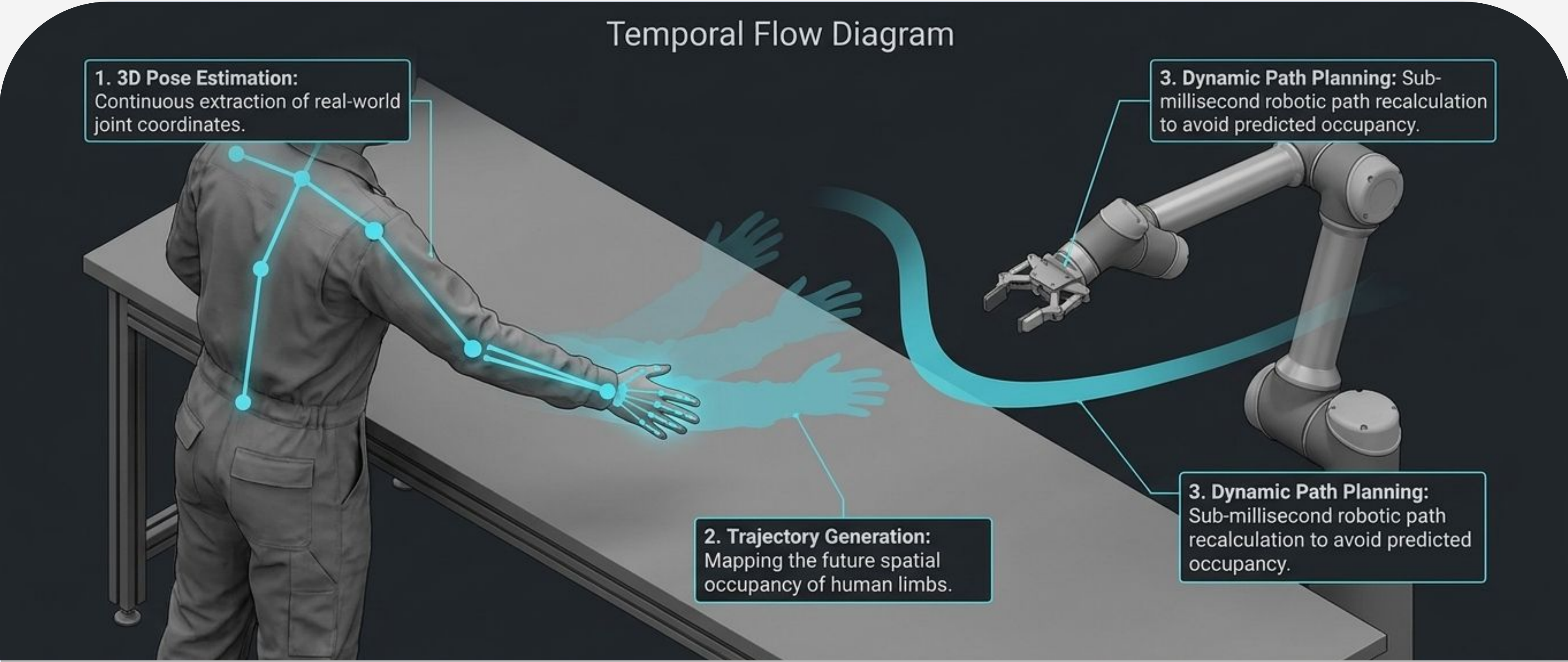
Technique 1: Reactive Safety via Speed and Separation Monitoring



Speed & Separation Monitoring (SSM)

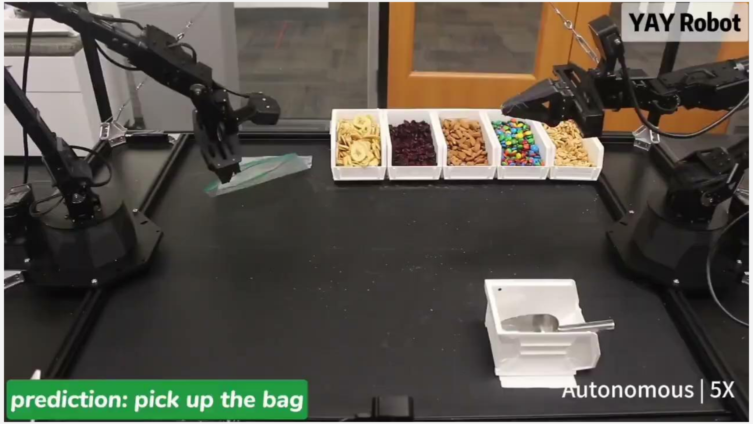
- **Purpose:** Prevent physical collisions via distance.
- **Yellow Zone:** Gradual deceleration (Warning). **Red Zone:** Emergency category 0 stop (Safety Halt).

Technique 2: Proactive Collision Avoidance



Insight: Shifting from bulk distance to kinematic intend protects physical integrity while preserving continuous, uninterrupted manufacturing flow.

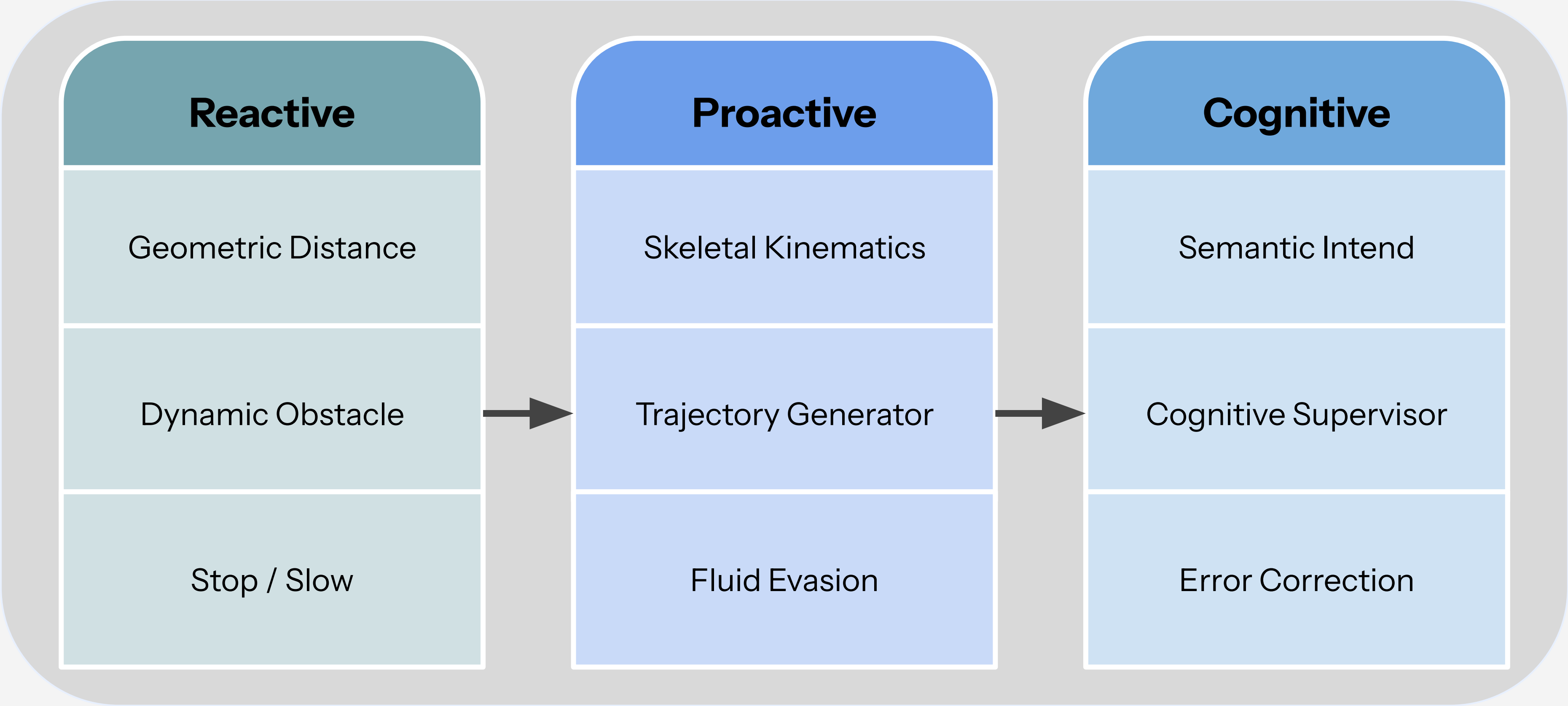
Technique 3: Cognitive Repair



Source: yay-robot.github.io/

Source: An Attention Transfer Model for Human-Assisted Failure Avoidance in Robot Manipulations

Evolution of Collaborative Robotics



Video of Vision 60 during fire fighting mission drill.
(Ghost Robotics. 2026. GR Corporate. *Vimeo*. vimeo.com/826958096/d1c124dbbb)

How **verbal communication** **is** used for corrections?

Alwin (DLR Institute of Robotic, 2025)

Voice-controlled robot solutions

Robots can understand and respond to human voice commands using AI technologies.

- **Input Module**

  Handles voice input and communication between humans and robots.
- **Hardware Module**

  Includes microphones, sensors, processors, and robotic actuators.
- **AI Control Module**

  Processes speech recognition, language understanding, and robot control.

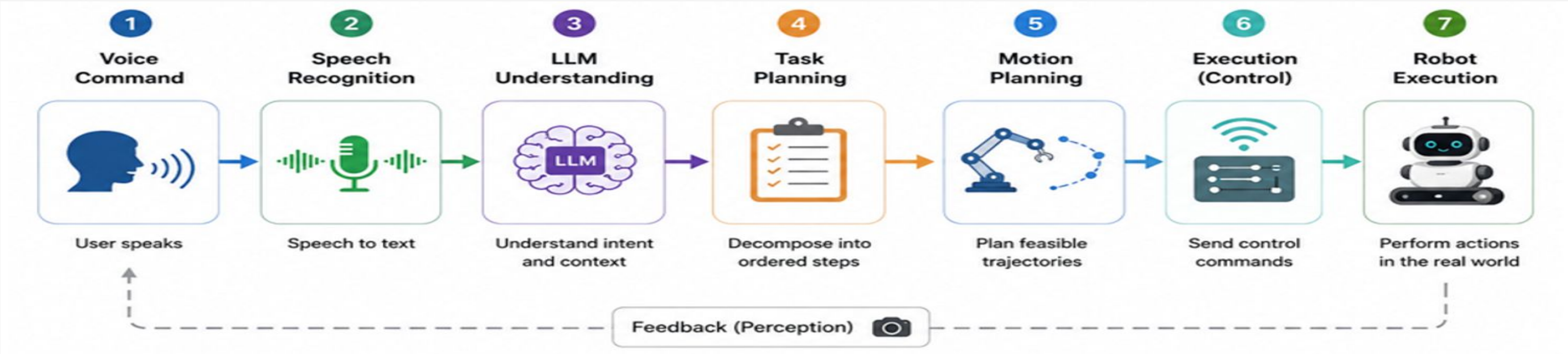
Common Methods for Voice-Controlled Robots

No	Method	Function
1	Speech Recognition (ASR)	Convert speech to text
2	NLP	Understand user commands
3	LSTM / RNN	Process voice sequences
4	Transformer / LLM	Advanced language understanding
5	Reinforcement Learning	Learn actions through interaction

- RNN and LSTM are older AI models. Today, Transformer has become the mainstream technology.
- LLM is a modern NLP technology based on Transformer architecture. NLP is a broader field, while LLM is one of its latest developments.

LLM-Driven Task and Motion Planning

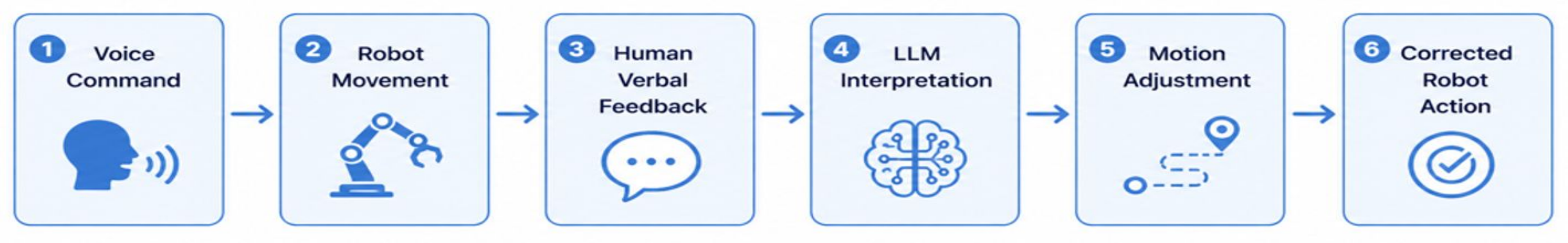
1. The robot sends voice commands to an LLM for understanding and task planning.
2. The LLM converts complex instructions into a sequence of executable actions.
3. The robot then performs these actions using motor control and computer vision.



Verbal Feedback-Based Robot Correction

- Human verbal feedback is interpreted by the LLM to correct robot actions in real time.
- The robot continuously updates its motion according to natural language instructions and corrections.

Examples: “Pick the red cup “ “Stop“ ”Move left”



Video of Digit taking a mug giving from human.
(Agility. 2026. Innovation at Agility: Five Fingered Hands. *YouTube*. https://youtu.be/w5iW60Q526k)

How **haptic** is used to reduce harm?

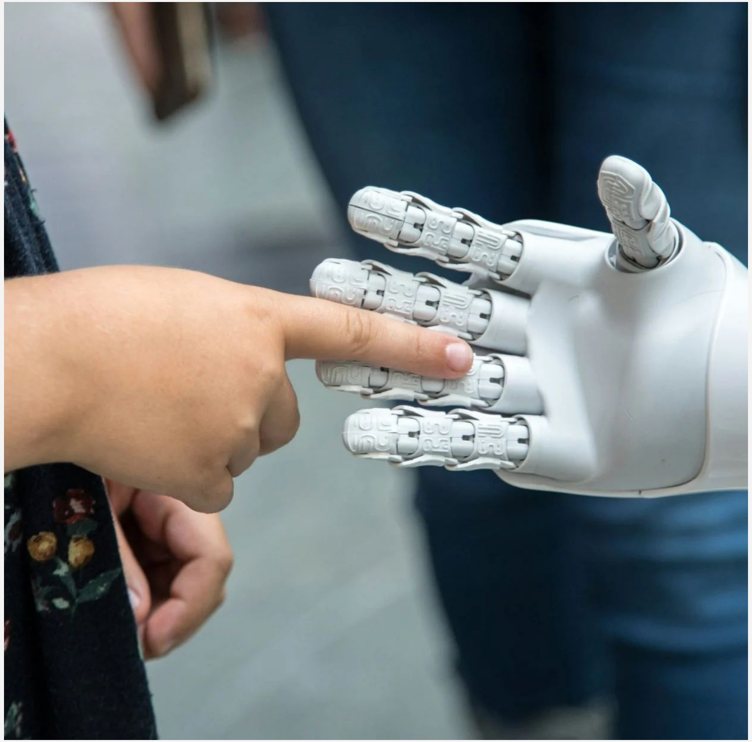
Hunter

(Aidin Robotics, 2024)

THE ULTIMATE SAFETY LINE

Haptic sensing provides the critical post-contact safeguard in Human-Robot Collaboration.

- Vision: Predictive avoidance (Pre-contact)
- Verbal: Error correction (During deviation)
- Haptic: Damage limitation (The Final Defense)



DECODING HAPTIC SENSING MODALITIES

Technology	Sensing Target	Key Application
Force / Torque Sensor	End-effector Forces	Collision Detection & Pushing
Joint Torque Sensor	External Joint Torque	Global Impact Localization
Tactile Robot Skin	Contact Pressure/Point	Multi-point Proximity Sensitivity
Motor Current Est.	Inferred Resistance	Cost-effective Impact Detection

Haptic sensors allow robots to **"feel"** rather than just "see" their environment.

THE 4-STAGE HARM REDUCTION WORKFLOW

Collision Detection

Instantaneous sensing of physical impact occurrence.

Collision Isolation

Pinpointing the exact contact point on the structure.

Force Identification

Estimating impact magnitude and direction.

Safe Reaction

Executing compliant control or a safety-rated stop.

BEYOND SAFETY: CULTIVATING TRUST

Converting Safety into Trust

Reliable haptic feedback bridges the gap between

Functional Safety and **Psychological Confidence**.

- Predictable responses build operator reliance.
- Immediate contact-stop prevents secondary injury.
- Perceived protection leads to higher efficiency.

How combined techniques can increase trust?

張泳德

(Boston Dynamics, 2024)

The Foundation of Human-Robot Collaboration

Trust is the cornerstone of any successful human-robot system. When trust is high, humans collaborate closely — with confidence and fewer interruptions.

The Problem

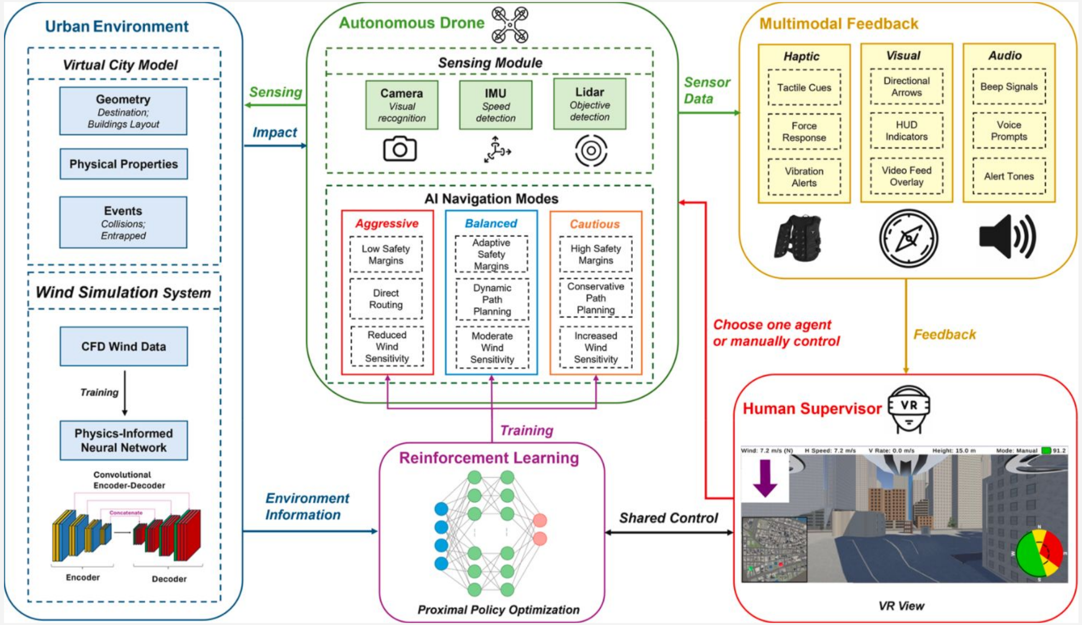
Real-world errors break trust, disrupting human-robot workflows

The Goal

Design systems that repair and proactively sustain trust after failures

Optimizing Trust in Autonomous Drones

Trust is equally shaped by **how AI communicates back to the human**. Research in drone shared autonomy reveals that sensory modality combinations dramatically affect workload, reliance, and collaboration quality.



Jiahao Wu, Bowen Sun, Tianyu Zhou, Fang Xu, Jing Du, Enhancing human-AI trust through multimodal feedback: A study of visual, haptic, and auditory augmentation in drone shared autonomy, Computers in Human Behavior Reports, Volume 21, 2026,

Optimizing Trust in Autonomous Drones

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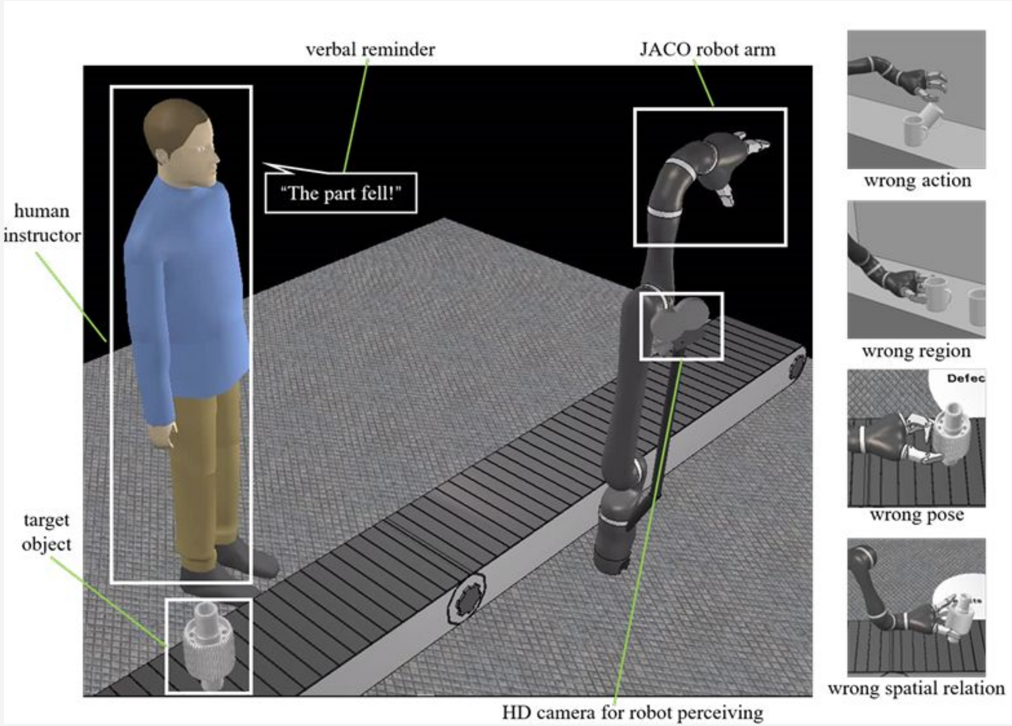
Modality Combination	Trust Level	Workload
Auditory Only	Low	High
Visual + Haptic	Highest	Lowest
Visual + Audio + Haptic	Highest	Moderate

Eye-tracking data specifically confirms that **haptic feedback reduces cognitive load**

Jiahao Wu, Bowen Sun, Tianyu Zhou, Fang Xu, Jing Du, Enhancing human-AI trust through multimodal feedback: A study of visual, haptic, and auditory augmentation in drone shared autonomy, Computers in Human Behavior Reports, Volume 21, 2026,

Repairing Trust with the H2R-AT Model



What it is?

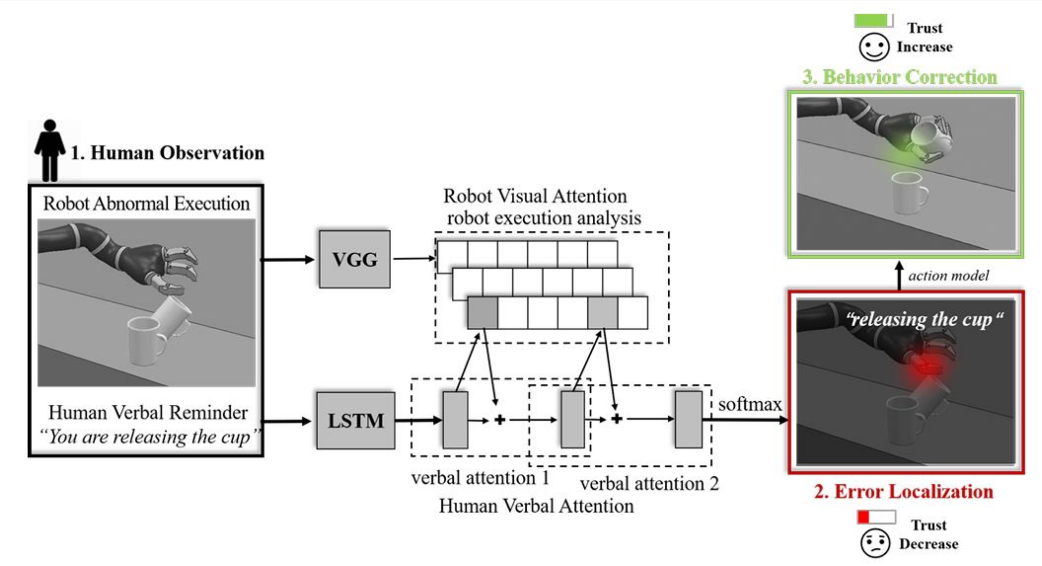
- Trust-maintenance framework
- Focuses on the psychological aspect of collaboration—trust

Working Sequences:

- Verbal Correction
- Visual focus
- Pose Correction

R. Luo, C. Huang, Y. Peng, B. Song and R. Liu, "Repairing Human Trust by Promptly Correcting Robot Mistakes with An Attention Transfer Model," 2021 IEEE 17th International Conference on Automation Science and Engineering (CASE), Lyon, France, 2021,

Repairing Trust with the H2R-AT Model



How it works?

- Combines natural language processing (NLP) and computer vision
- LSTM network: interpret the human's verbal instructions
- VGG: Extract feature vectors from camera
- Attention Transfer: transfer human's verbal focus

R. Luo, C. Huang, Y. Peng, B. Song and R. Liu, "Repairing Human Trust by Promptly Correcting Robot Mistakes with An Attention Transfer Model," 2021 IEEE 17th International Conference on Automation Science and Engineering (CASE), Lyon, France, 2021,

Statistical Proof of Trust Repair

A user study with **252 participants** validated the H2R-AT model across all four error categories.

90.75%
Decision Reliability
Probability of the robot choosing the correct solution

73.68%
Attention Accuracy
Human verbal attention correctly transferred to the robot

66.86%
Failure Avoidance
Robot failure avoidance guided by human verbal input

73%
Trust Restored
Trust recovered to pre-mistake baseline level

- Robot mistakes can be repaired by a prompt by verbal attention to robot’s visual focus

R. Luo, C. Huang, Y. Peng, B. Song and R. Liu, "Repairing Human Trust by Promptly Correcting Robot Mistakes with An Attention Transfer Model," 2021 IEEE 17th International Conference on Automation Science and Engineering (CASE), Lyon, France, 2021,

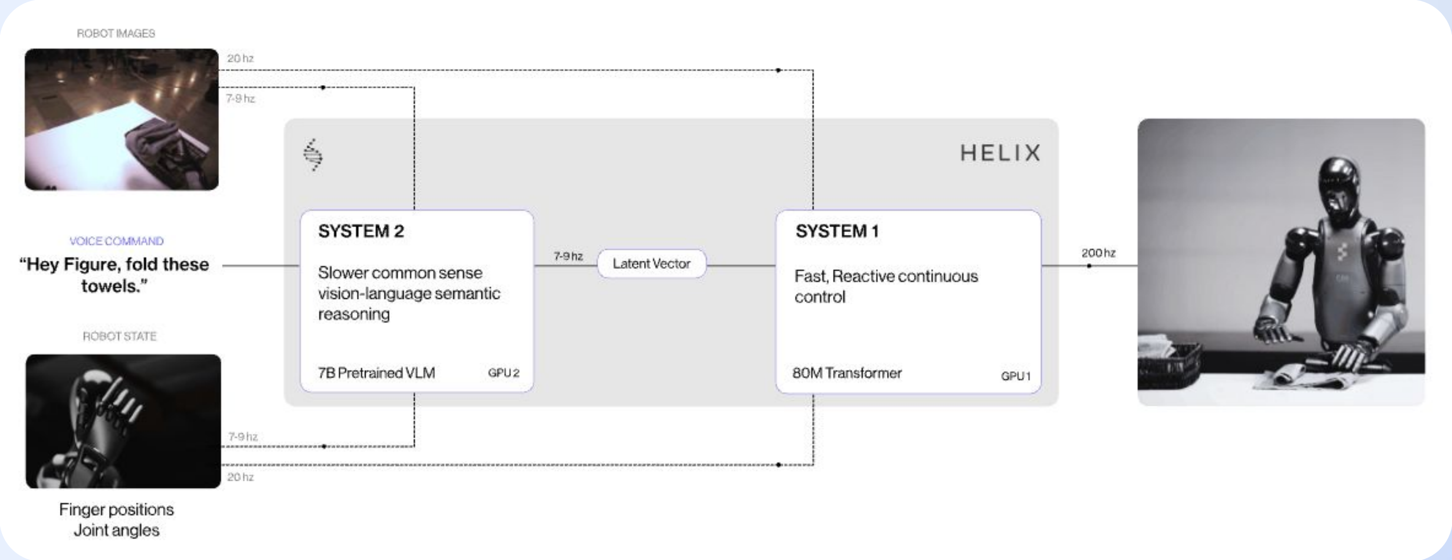
Take Away

Henry

(Boston Dynamics, 2026)

What you have learned?

(Figure AI, 2025)



01

Vision allow cognitive understanding

02

Verbal is nature way of **Communication**

03

Physical limits serviced as last line of defence

04

Multi-modals and corrections increase **Trust**

Q&A

Video of Atlas back heel kick soccer ball.
(Boston Dynamic. 2026. School of Football | The Rabona | Boston Dynamics x Hyundai. *YouTube*. https://youtu.be/mCcv90sUNbY)

Reflections

The field of human-robot collaboration (HRC) is broad. Initially, the team came up with a vast variety of topics from RL for fleet-scale robot policies, to real-time transformer networks for human intention recognition, and to vision-language-action (RT-2) transfer models. It was a hard pick, so by the process of voting, we ended up on Repairing Human Trust by Promptly Correcting Robot Mistakes with an Attention Transfer Model (H2R-AT), and narrowed down the topic to focus on the safety and trust aspects in HRC. Still, it is a broad area of research, and hardly covered within a 20-minute speech, so each of the teammates had rehearsed and condensed the content again and again. Overall, we think the coverage for each sub-topic (vision, verbal, haptic) is general and relevant enough to lead to the main topic of repairing trust. If we have enough time, we think we could survey for more video footage on the latest studies or applications in industries for better connections to emphasize the importance of trust and how it will impact our society in the future.

References

Luo, Ruijiao, et al. "Repairing human trust by promptly correcting robot mistakes with an attention transfer model." *2021 IEEE 17th International Conference on Automation Science and Engineering (CASE)*. IEEE, 2021.

Mavsar, Matija, Mihael Simonič, and Aleš Ude. "Human intention recognition by deep LSTM and transformer networks for real-time human-robot collaboration." *Frontiers in Robotics and AI* 12 (2025): 1708987.

Robinson, Nicole, et al. "Robotic vision for human-robot interaction and collaboration: A survey and systematic review." *ACM Transactions on Human-Robot Interaction* 12.1 (2023): 1-66.

Shahriari, Mohammadali, and Mohammad Biglarbegian. "A novel predictive safety criteria for robust collision avoidance of autonomous robots." *IEEE/ASME Transactions on Mechatronics* 27.5 (2021): 3773-3783.

Shi, Lucy Xiaoyang, et al. "Yell at your robot: Improving on-the-fly from language corrections." *arXiv preprint arXiv:2403.12910* (2024).

Wu, Jiahao, et al. "Enhancing human-AI trust through multimodal feedback: A study of visual, haptic, and auditory augmentation in drone shared autonomy." *Computers in Human Behavior Reports* 21 (2026): 100959.

Zitkovich, Brianna, et al. "Rt-2: Vision-language-action models transfer web knowledge to robotic control." *Conference on Robot Learning.* PMLR, 2023.

Credits

Members	Roles
314540056 格若苿 (Gaurav Meena)	Introduction slides (p.3-7)
314540072 鄭柏維 (Panuwat Saetee)	Topic on Vision (p.10-14)
412551033 楊錦程 (Alwin)	Topic on Verbal Communication (p.17-20)
414551016 劉順松 (Hunter)	Topic on Haptic/Physical sensors (writer; p.23-26)
314540006 張泳德	Topic on combined technique (p.28-32)
314553047 吳普軒 (Henry)	Team coordinator, Speaker for Haptic slides, Remaining slides